

Abstract

This paper is an introduction to beanPy: a python library that generates samples of random numbers based on various statistical distributions. This paper includes information on which distributions beanPy contains, the functions of beanPy and an example of where beanPy can be used.

An Innovative Random Number Generator: BeanPy

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1 Introduction

The normal distribution [2], the exponential distribution [1], the Poisson distribution [6] and the continuous and discrete uniform distributions [4] [5] are all fundamental distributions well known to statisticians. What beanPy aims to do is utilise these distributions and turn them into random number generators that depend on their own specific parameters. Standard random number generators choose a number between zero and one with each outcome having equal probability, i.e., these random number generators use a uniform distribution to find random numbers. beanPy is a little different, however. beanPy allows you to generate random numbers with differing probabilities, and these randomly generated numbers are not restricted to between zero and one.

beanPy works by taking a sample from a population that follows a given distribution. For each of the distributions that beanPy works with, we can choose any required parameters associated with each distribution. For example, if we are using the normal distribution, we can choose its mean and variance so that we can achieve our required probabilities. Also, for each distribution we can choose a seed so that we get the same probability each time.

Furthermore, for each distribution we can graph its probability density function (pdf) and its cumulative distribution function (cdf) associated with its given parameters. This is done using the matplotlib.pyplot library to draw the graphs.

2 Statement of need

beanPy could be useful in many different scenarios. Consider the case where someone is throwing a dart at a target while aiming for ‘triple twenty’ [3]. Just because said person is aiming for triple twenty it doesn’t mean that they’re going to hit it every time. But since they are aiming for triple twenty, it is more likely that their efforts will concentrate around where they are aiming. So, in this case the probabilities of hitting any point on the target when you aim for triple twenty is more likely to be normally distributed rather than uniformly distributed. Hence using the normal distribution would be more appropriate to achieve more accurate probabilities of hitting a target at any given point. This is assuming that we have a skilled darts player that doesn’t throw a dart at random.

Here is how we can use beanPy to model a dart player using a bivariate normal distribution:

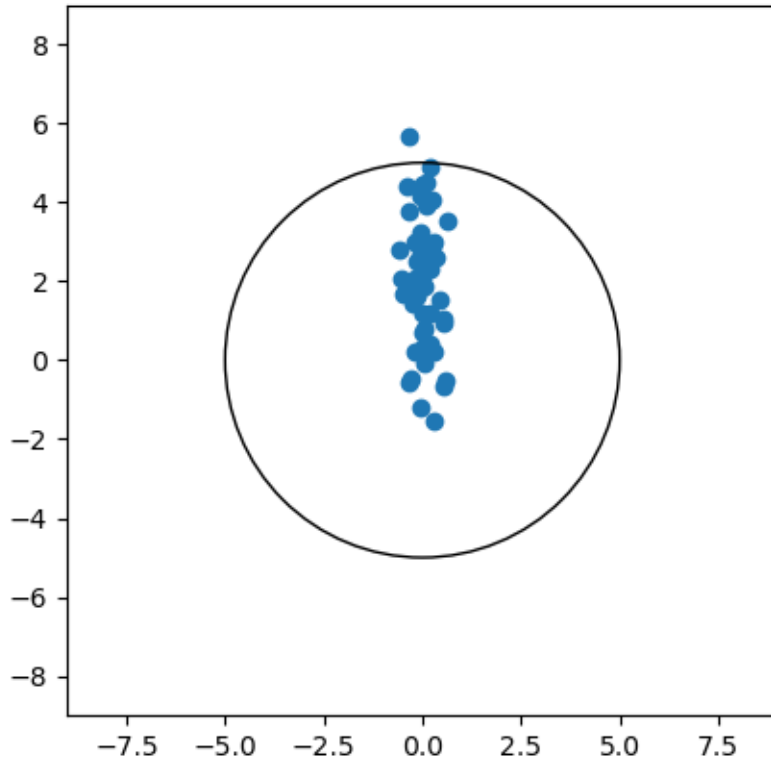
```
import beanPy as Beans
import matplotlib.pyplot as plt

dart_player_horizontal = Beans.NormalDistribution(0, 0.1)
dart_player_vertical = Beans.NormalDistribution(2, 4)

number_of_darts = 50
x_values = dart_player_horizontal.TakeMultipleSamples(number_of_darts, 1)
y_values = dart_player_vertical.TakeMultipleSamples(number_of_darts, 2)

figure, axes = plt.subplots() #Generates our blank graph
draw_dartboard = plt.Circle((0,0), 5, fill=False) #Adds our dartboard
plt.xlim([-9, 9]) #Sets our x axis
plt.ylim([-9, 9]) #Sets our y axis
axes.set_aspect(1) #Keeps the plot in a 1:1 ratio
axes.add_artist(draw_dartboard) #Adds our dartboard to the plot
darts = plt.scatter(x_values, y_values) #Adds our darts to the plot
plt.show() #Shows the plot
```

This is the graph that our code written above generates:



In this example the darts player is very accurate with his horizontal aim, but is struggling with his vertical aim leading to the majority of his shots over/under shooting, but rarely being far to the side. This scenario could be caused by the darts player being unsure about the distance to the dart board as this would only affect his ability to aim on the y-axis but not the x-axis, causing this distribution of shots.

3 Conclusion

This paper has given a description of the python library beanPy. The aim of the beanPy library is to generate random samples of numbers based off a few statistical distributions. This library is limited as it only includes the normal, exponential, Poisson, and both continuous and discrete uniform distributions however many more distributions could be added in the future as the code has been written in a modular way such as to allow new distributions to be added which use the same functions as the other distributions. Also, the functionality of being able to draw separate graphs on the same axis could be implemented in the future so that graphs can be more easily compared.

References

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